

VARENNES, QC, Canada

WMO: 711840

Lat: 45.7229N

Lon: 73.3765W

Elev: 59

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.4	-5.9	-19.7	1.9	-9.1	-15.1	2.4	-5.1	31.9	29.2	27.9	22.9	8.6	270	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	86.5	71.2	83.6	69.6	80.9	68.1	74.2	82.6	72.4	80.0	70.6	77.8	10.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	116.4	78.9	69.8	109.7	76.8	68.1	103.4	74.9	37.8	82.5	36.1	80.0	34.6	77.7	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	21.2	18.9	DB	-17.3	90.9	6.1	2.6	-21.7	92.8	-25.3	94.4	-28.7	95.9	-33.2	97.8	(4)
(5)			WB	-17.7	77.8	6.0	2.2	-22.1	79.4	-25.6	80.6	-29.0	81.8	-33.3	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	14.9	17.3	28.0	42.6	56.5	65.5	69.1	67.3	59.8	47.6	35.8	23.4		
	DBStd	20.93	12.46	10.85	10.47	8.00	7.61	6.62	5.04	5.25	7.05	7.72	8.58	10.97		
	HDD50	4362	1089	915	685	247	23	0	0	0	6	142	429	826		
	HDD65	8012	1554	1335	1148	674	282	73	17	36	185	542	875	1291		
	CDD50	2213	0	0	2	24	223	465	592	536	301	66	4	0		
	CDD65	390	0	0	0	1	18	88	144	108	30	1	0	0		
	CDH74	3170	0	0	1	17	246	763	1103	799	234	7	0	0		
	CDH80	765	0	0	0	3	60	213	267	172	50	0	0	0		
	WSAvg	8.6	10.4	10.1	9.9	9.8	9.3	7.7	6.8	6.1	6.7	8.0	9.1	9.7		
	PrecAvg	40.2	3.0	2.3	2.6	3.3	3.5	3.9	4.0	3.8	3.5	3.9	3.0	3.5		
	PrecMax	49.8	5.6	4.1	4.7	6.0	7.2	7.7	6.2	7.0	7.8	7.2	5.5	6.5		
(15) Precipitation	PrecMin	32.4	1.1	0.9	0.7	0.7	1.7	1.8	2.1	1.5	1.0	1.2	0.5	1.1		
	PrecStd	4.3	1.3	0.8	1.0	1.5	1.3	1.4	1.0	1.3	1.5	1.6	1.2	1.4		
	DB	44.1	41.5	58.2	75.9	86.5	89.8	90.0	88.0	86.1	74.5	62.1	51.1			
	MCWB	42.9	38.3	49.4	59.3	65.9	72.0	75.1	73.1	71.3	62.8	57.1	48.1			
	2% DB	39.3	38.2	49.8	67.2	80.5	85.6	85.7	84.4	80.0	68.7	56.6	44.6			
(21) Mean Coincident Wet Bulb Temperatures	MCWB	37.0	35.7	43.4	52.6	61.7	69.1	72.0	71.1	68.4	59.6	52.7	42.4			
	5% DB	36.1	35.9	44.1	61.1	75.8	82.0	82.8	81.4	76.1	63.6	52.1	40.4			
	MCWB	34.5	33.8	39.5	49.5	59.6	67.3	70.6	69.5	66.0	56.6	48.2	38.3			
	10% DB	33.0	33.9	40.6	56.0	71.0	78.4	80.1	78.5	72.4	59.3	48.1	36.9			
	MCWB	31.6	32.3	37.0	46.4	58.0	65.3	69.0	68.3	63.9	53.6	44.5	35.0			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	42.6	39.0	51.9	61.6	70.1	74.7	77.3	76.2	73.7	65.3	58.7	48.5			
	MCDB	43.6	40.3	55.4	72.8	81.7	86.0	87.6	84.5	82.0	71.3	61.0	51.0			
	2% WB	37.4	36.0	43.9	55.3	66.3	72.2	74.7	73.4	70.3	60.8	53.1	42.5			
	MCDB	38.7	37.6	48.8	62.7	74.1	81.8	82.2	81.0	77.2	67.1	56.3	44.2			
	5% WB	34.6	34.2	40.0	50.7	63.0	70.0	72.7	71.4	67.8	57.5	48.6	38.4			
(31) Mean Daily Temperature Range	MCDB	36.0	35.7	43.3	58.6	71.6	78.2	79.9	78.3	73.9	62.3	51.5	40.0			
	10% WB	31.6	32.3	37.3	47.5	59.9	67.9	70.9	69.6	65.3	54.6	45.0	35.3			
	MCDB	32.9	33.9	40.4	54.7	67.6	74.8	77.7	76.2	70.9	58.4	47.7	36.6			
	5% DB	14.8	15.8	15.3	18.1	20.3	18.8	18.6	19.5	19.9	16.0	12.9	12.4			
	MCDBR	15.7	14.6	17.9	28.2	27.9	24.3	21.8	22.5	23.5	22.4	18.9	13.5			
(36) Clear-Sky Solar Irradiance	MCWBR	15.2	13.4	13.2	16.4	13.7	11.7	11.0	11.7	13.1	14.0	15.6	12.5			
	5% WB	15.5	13.9	17.0	24.4	22.4	19.8	18.7	20.0	20.5	19.4	18.1	13.0			
	MCWBR	15.1	13.4	13.2	16.2	13.1	11.4	11.0	11.5	12.7	13.4	15.8	12.4			
	taub	0.256	0.290	0.337	0.365	0.394	0.438	0.429	0.416	0.376	0.337	0.308	0.297			
	taud	2.352	2.241	2.249	2.341	2.322	2.205	2.234	2.271	2.369	2.485	2.481	2.284			
(42) All-Sky Solar Radiation	Ebn at Noon	274	281	281	283	277	264	265	264	266	263	252	238			
	Edh at Noon	23	32	37	37	39	44	43	40	33	25	21	23			
	RadAvg	457	754	1150	1392	1624	1785	1785	1576	1245	720	434	314			
(45)	RadStd	54	80	96	157	157	134	138	73	102	62	56	27			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page